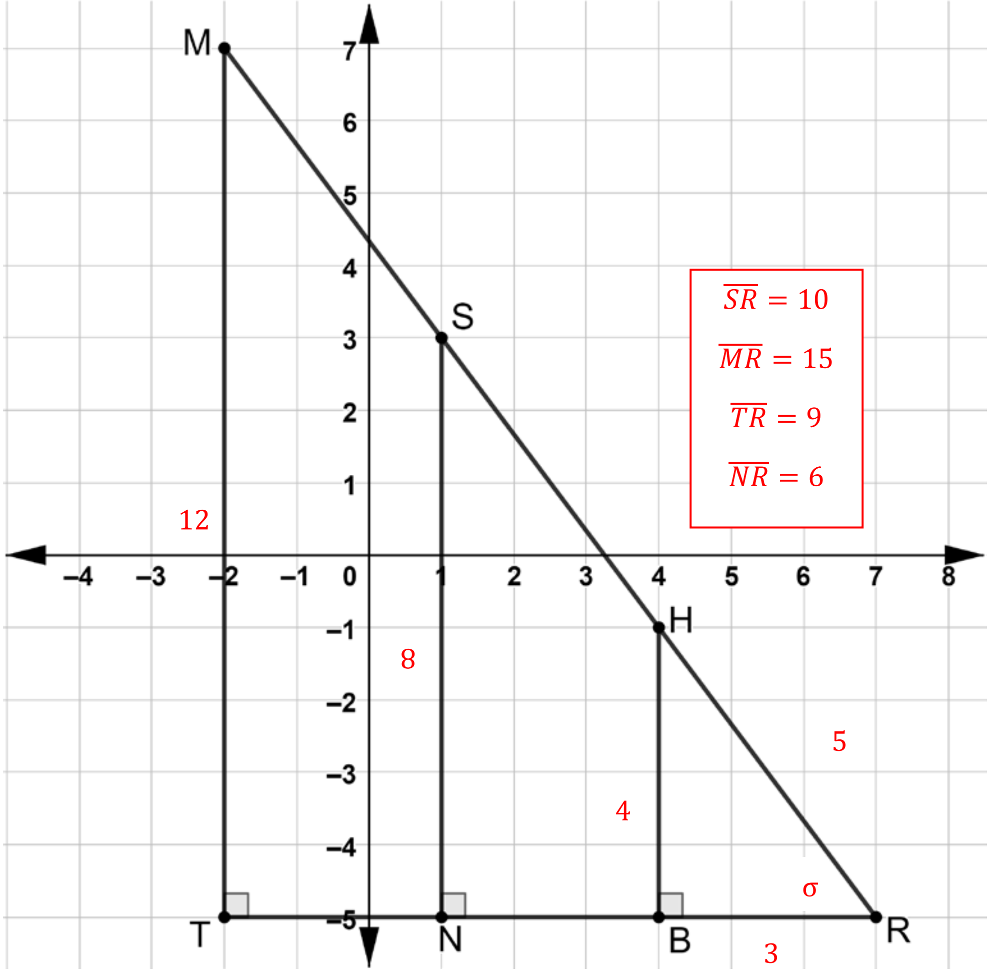
Geometry Name_Answer Key________
Unit 7 Date____________________
Lesson 4 Practice Period__________________

Three right triangles are shown on the coordinate grid.



The coordinates of the vertices are shown.

$M\ (-2,7)$ $S\ (1,3)$ $H\ (4,-1)$ $R\ (7,-5)$

$T\ (-2,-5)$ $N\ (1,-5)$ $B\ (4,-5)$

1. Explain why these triangles are similar.

All three triangles share $\angle R$ and they all have a right angle.

By AA similarity, they are similar. In addition, the sides are proportional.

$$\frac{12}{9} = \frac{8}{6} = \frac{4}{3}$$

Complete the table.

	Triangle	Side	Length
2.	HRB	$\overline{BH}$	4
		$\overline{BR}$	3
		$\overline{RH}$	5 $3^2 + 4^2 = c^2$
3.	SRN	$\overline{NS}$	8
		$\overline{RN}$	6
		$\overline{SR}$	10 $6^2 + 8^2 = 10^2$
4.	MTR	$\overline{MT}$	12
		$\overline{RT}$	9
		$\overline{MR}$	15 $12^2 + 9^2 = c^2$

Using $\angle R$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

	Triangle	Cosine	Sine	Tangent
5.	HRB	$\frac{A}{H} = \frac{3}{5} = 0.6$	$\frac{O}{H} = \frac{4}{5} = 0.8$	$\frac{O}{A} = \frac{4}{3} = 1.\overline{3}$
6.	SRN	$\frac{A}{H} = \frac{6}{10} = 0.6$	$\frac{O}{H} = \frac{8}{10} = 0.8$	$\frac{O}{A} = \frac{8}{6} = 1.\overline{3}$
7.	MTR	$\frac{A}{H} = \frac{9}{15} = 0.6$	$\frac{O}{H} = \frac{12}{15} = 0.8$	$\frac{O}{A} = \frac{12}{9} = 1.\overline{3}$

Use your calculator to find each. Round to the nearest thousandth.

	Trigonometric Ratio	Value rounded to nearest thousandth
8.	$cos(53°)$	0.602
9.	$sin(53°)$	0.799
10.	$tan(53°)$	1.327